Department of Computer Science and Engineering

Academic Year 2021– 2022(Odd Semester)

Degree, Semester & Branch: B.E – I Semester Mech

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member (s): Mrs.M.Dhivya, AP/CSE

Innovative Practice Description

- **Unit / Topic: Unit IV / Dictionaries operation and method**
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 11 12

Activity Chosen: MindMap

- **Justification:**
  A mind map is a visual representation that shows the concepts and ideas. It's a technique for visual thinking that helps with data arrangement, so this practice will help to student recognize the theoretical ideas in the image and facilitate quick recall for their tests.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
  - Faculty explained the concept of Dictionary operations and methods.
  - Based on the discussion the teacher asked the Student to draw a mind map related to the topic within 15 minutes.
  - Each student created a mind map on Dictionaries: operation and methods.
- The faculty member collected the sheet and appreciated the students.

CO – PO / PSO mapping:

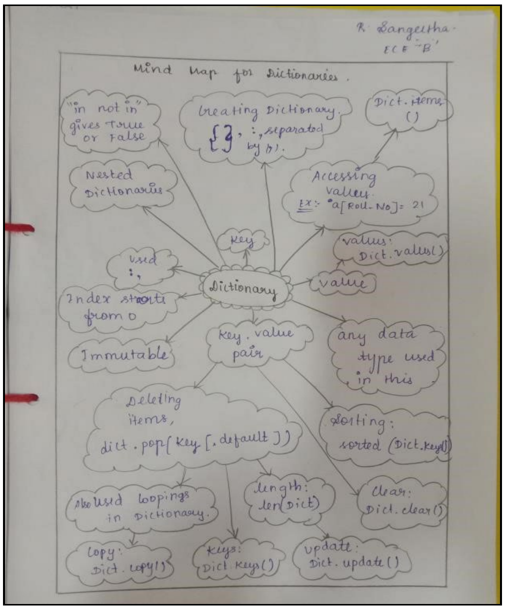
CO	PO1	PO2	PO3	PO5	PO9	PSO1
CO4	3	1	1	1	1	1

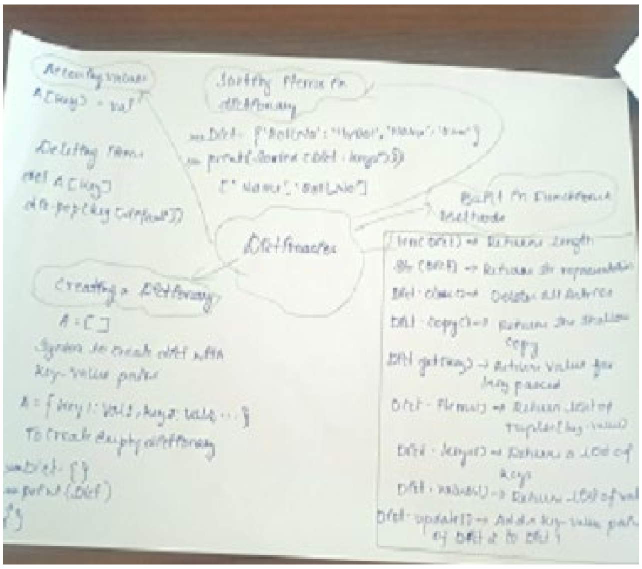
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**Justification for correlation**
PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO5	Modern tool usage**:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO9	Project management and finance**:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PSO1	Students will be able to provide sustainable solution to mechanical engineering problem using the various algorithmic tools.

- **Images / Screenshot of the practice:**





- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
  - Students expressed that they understood the concept very well and the pictorial representation made them to understand the topic clearly.
  - This helped the students in recalling the topic taught on that specific day, generating new ideas about the topic.
  - ***Benefit of the practice:***
    - From this activity, the students can get more clarity in the particular topic.
    - Through this activity, the students are able to remember and describe the various dictionary operations and it methods clearly.
  - C***hallenges faced in implementation:***
    - Some students missed some few methods in writing the mind map.
    - Some of the students feel difficult to depict the concepts pictorially.
    - Some of them need some more clarification on the dictionary operations. Additional examples given to make them to understand the concept easily.

References:

- https://www.slideshare.net/raja2dlas/minute-paper
- https://www.ritrjpm.ac.in/images/computer-science/CS8251-SE-MinutePaper.pdf